

Calibrated, Leakage-Safe Ranking of Short-Horizon Equity Setups: An Honest Walk-Forward Study on NSE Stocks

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Abstract

Retail equity-screening systems often report attractive backtests while hiding lookahead leakage, weak calibration, or overfit selection rules. This paper presents a modest empirical study of a reproducible pipeline for ranking short-horizon equity setups on Indian National Stock Exchange (NSE) stocks. For each stock and trading day, the system constructs causal technical, relative strength, market-breadth, sector, regime, and earnings-proximity features using only information available up to the close of day t . A concrete trade is then defined using an ATR-based entry band, target, stop, and approximately five-day holding horizon. The label is produced by a triple-barrier rule: target first is a win, stop first is a loss, and unresolved trades exit at the horizon. We train a gradient-boosted tree classifier to estimate the probability that a setup hits target before stop, calibrate this probability on a time-ordered holdout tail, and evaluate all predictions using walk-forward testing with an embargo equal to the holding period. The contribution is not a claim of a profitable trading system. Instead, we ask whether calibrated, leakage-safe ranking improves over the unranked base rate and rule-based ranking when measured honestly out of sample. All performance numbers in this draft are placeholders until the experiment scripts are run and independently checked.

Keywords: financial machine learning; probability calibration; walk-forward validation; triple-barrier labeling; NSE equities; leakage-safe backtesting.

Suggested arXiv categories: cs.LG, q-fin.ST, cs.CE.

1 Introduction

Short-horizon stock-selection tools are attractive because they convert noisy market data into an apparently simple ranked list of candidates. The same simplicity also makes them easy to overstate. A trading model can look useful if it is evaluated on future information, tuned repeatedly on the same period, or judged only by raw accuracy rather than by calibrated probabilities and realistic trade outcomes. These failure modes are especially important for student or retail research, where the goal should be reproducibility and honest measurement rather than promotional claims.

This paper studies a single question:

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Does calibrated, leakage-safe, walk-forward ranking of NSE equity setups beat the un-ranked base rate and a rule-based ranking baseline out of sample, and by how much when the trades are graded in R-multiples after costs?

The system under study is a daily pipeline that defines a concrete short-horizon trade for each stock-day and estimates the probability that the trade reaches its target before its stop. The model is not evaluated by whether it can predict the next close exactly. It is evaluated by whether the ranked setups produce better out-of-sample barrier outcomes than simpler baselines.

The intended contribution is deliberately modest:

1. We describe a leakage-safe data pipeline that creates causal stock-day features and triple-barrier labels for short-horizon NSE equity setups.
2. We estimate and calibrate setup win probabilities using a time-ordered training and calibration design rather than random cross-validation.
3. We report an honest walk-forward evaluation, including classification metrics, probability reliability, economic R-multiple metrics, baselines, ablations, sensitivity checks, and limitations.

This work should not be read as financial advice or as evidence that the system is safe for live trading. A realistic outcome for this type of model may be only a thin edge, no edge, or an edge concentrated in specific market regimes. That is precisely why the evaluation protocol is the main object of the paper.

2 Related Work

Machine learning has been applied widely to equity prediction and asset pricing. Large-scale studies such as Gu et al. [2] show that non-linear models can extract predictive structure from financial characteristics, but also that evaluation design matters. Technical-indicator based equity prediction is a smaller and noisier setting: many candidate signals are correlated, non-stationary, and sensitive to transaction costs.

This study follows several ideas from financial machine learning practice. Lopez de Prado [1] emphasizes meta-labeling, triple-barrier labeling, and purged or embargoed validation to avoid overlap between training labels and test periods. Backtest overfitting is a known risk in strategy research [10, 11]; therefore, this paper treats parameter sweeps as sensitivity analysis rather than proof.

Probability calibration is also central. A model can rank examples reasonably while producing probabilities that do not match realized frequencies. Platt [3] introduced a widely used sigmoid calibration approach for support vector machines, while Niculescu-Mizil and Caruana [4] compare calibration methods such as Platt scaling and isotonic regression. We use isotonic calibration because the desired output is an interpretable estimate of setup win probability, not only a ranking score. Calibration is evaluated using the Brier score [5] and reliability tables. Discrimination is summarized using ROC-AUC [6].

The model family is gradient-boosted decision trees, implemented using the histogram-based gradient boosting classifier provided by scikit-learn [7]. Gradient boosting is a strong baseline for tabular data with non-linear interactions; related systems include LightGBM [8]. The contribution here is not a new model architecture. It is the construction, calibration, and evaluation of a transparent empirical pipeline for a specific stock-selection task.

3 Data

3.1 Source and Universe

The dataset is built from daily open, high, low, close, and volume (OHLCV) data for Indian NSE equities, downloaded or cached through `yfinance` [13]. The market context uses the Nifty 50 index (^NSEI). The current project uses approximately 474 NSE stocks for training and a smaller large-cap daily-signal universe of approximately 80 stocks for deployment-style signal generation.

The approximate training table contains 491,000 stock-day rows over about 1,125 trading days. The unranked base win rate is approximately 31.6%. Exact values should be regenerated from the experiment scripts and inserted before submission: `[[RESULT: exact dataset rows]]`, `[[RESULT: exact symbol count]]`, `[[RESULT: exact date range]]`, and `[[RESULT: exact base win rate]]`.

3.2 Stock-Day Row Construction

For every stock i and trading day t , the pipeline creates one candidate row using only information available up to the close of day t . Features are computed from historical price, volume, index, sector, breadth, and earnings calendar data. The label is computed from future bars $t+1, \dots, t+H$, where H is the holding horizon. Features and labels are therefore separated in time: past data defines the input, and future data defines the outcome.

3.3 Preprocessing

Rows with insufficient rolling-window history are skipped. Indicators requiring 20-day, 50-day, or 14-day windows are computed only after enough observations exist. The index series is aligned by trading date. The system uses conservative handling for ambiguous daily bars because daily OHLCV data cannot tell whether the target or stop was hit first within the same session.

4 Methodology

4.1 Feature Set

Table 1 summarizes the causal feature groups. Each feature is computed as-of day t only.

4.2 Trade Definition and Triple-Barrier Label

For each row, the rule engine defines a direction, entry band, target, and stop using the latest close and average true range (ATR). For a buy setup, the target is above the entry and the stop is below it; for a sell setup, the directions are reversed. Let R denote the risk distance from entry to stop. Outcomes are expressed in R-multiples so trades with different prices and volatility can be compared.

The label is assigned over the next H trading sessions:

- **WIN** if the target is hit before the stop; the binary label is $y = 1$ and realized R is the target-to-risk multiple.
- **LOSS** if the stop is hit before the target; the binary label is $y = 0$ and realized R is -1 .
- **TIME_EXIT** if neither barrier is touched by the horizon; the binary label is conservatively treated as $y = 0$, while realized R is marked to the horizon close.

Table 1: Causal feature groups used for each stock-day row.

Group	Features and definition
Trend and momentum	Close relative to EMA20; EMA20 relative to EMA50; 5-day and 20-day returns; RSI(14); ADX(14).
Volatility and range	ATR as a percentage of close; daily high-low range as a percentage of close.
Volume and liquidity	Current volume divided by 20-day average volume; liquidity score used in final ranking.
Support and resistance	Distance from 20-day resistance in ATR units; distance from 20-day support in ATR units.
Relative and sector strength	Relative strength versus the market index; mean sector relative strength; sector breadth relative to market breadth.
Market regime and breadth	Regime score; buy/sell regime alignment; fraction of universe above EMA50 and EMA20; fraction advancing over five days; change in breadth.
Rule-engine context	Direction indicator; buy score; sell score; chart setup quality.
Earnings proximity	Days since earnings; clipped days until earnings; pre-earnings flag; post-earnings flag. Future earnings dates are clipped to reduce leakage from unscheduled future events.

If both target and stop are touched within the same daily bar, the default offline experiment scores the trade as a loss. This is pessimistic, but it avoids flattering the strategy when intraday ordering is unknown.

4.3 Model

The predictive model is a scikit-learn `HistGradientBoostingClassifier`. It is chosen because gradient-boosted trees are strong, CPU-efficient models for tabular data with non-linear interactions and missingness-resistant splits. The model estimates

$$\hat{p}_{i,t} = P(y_{i,t} = 1 \mid x_{i,t}),$$

where $x_{i,t}$ is the causal feature vector for stock i on day t .

The raw classifier output is calibrated with isotonic regression using `CalibratedClassifierCV`. Calibration is fit on the time-ordered tail of the training period, not on randomly shuffled folds. This preserves the temporal structure of the problem.

4.4 Expected R and Final Ranking

For a candidate setup, expected R is computed as

$$E[R] = p_{\text{win}} \cdot R_{\text{target}} - p_{\text{loss}} \cdot R_{\text{stop}} - c_{\text{txn}} - c_{\text{slip}},$$

where the default transaction and slippage costs are approximately $0.03R$ and $0.02R$, respectively.

The deployed ranking score is a weighted combination of calibrated win probability, expected R, historical setup win rate, regime alignment, chart quality, relative strength, liquidity, model

stability, and risk penalties:

$$\begin{aligned}\text{final_score} = & 0.30p_{\text{win}} + 0.20E[R] + 0.15\text{AdjSetupWinRate} \\ & + 0.10\text{RegimeAlignment} + 0.10\text{ChartQuality} + 0.05\text{RelativeStrength} \\ & + 0.05\text{Liquidity} + 0.05\text{ModelStability} - \text{RiskPenalties}.\end{aligned}$$

For the balanced profile, a signal is shown only if it clears gates such as calibrated $p_{\text{win}} \geq 0.64$, risk-reward ≥ 1.55 , positive expected R , and final score ≥ 0.58 . These deployment gates are evaluated separately from the broader ranking study. Diversification caps highly correlated selections, with pairwise recent-return correlation above 0.92 treated as a duplicate exposure.

4.5 Walk-Forward Validation With Embargo

All reported metrics are based on concatenated out-of-sample predictions from a walk-forward design. The default protocol is:

1. Sort all rows by trading date.
2. Use an initial warm-up period of approximately 500 trading days.
3. Train only on rows strictly before the test window.
4. Remove an embargo gap equal to the holding horizon ($H = 5$ days) so training labels cannot overlap the test window.
5. Fit calibration on the last 15% of the training dates.
6. Predict the next out-of-sample window, approximately 60 trading days.
7. Slide forward and repeat until all available out-of-sample windows are predicted.

This design is stricter than random cross-validation because no future row is allowed to train a past prediction.

5 Experiments and Results

5.1 Experiment Commands

The following commands regenerate the main experiment outputs from the project root after back-end environment variables and caches are configured:

```
cd backend
python scripts/train_ml.py --hold 5 --top-n 5 --cost 0.05
python scripts/train_ml.py --hold 5 --top-n 10 --cost 0.05
python scripts/analyze_ml_oos.py
python scripts/run_backtest.py --risk Balanced --top-n 5 --cost 0.05 --hold 5
python scripts/run_backtest_sweep.py
```

If caches need to be rebuilt, run:

```
cd backend
python scripts/train_ml.py --rebuild --hold 5 --top-n 5 --cost 0.05
```

5.2 Main Classification Metrics

Table 2 reports out-of-sample classification quality. These values must be filled from `scripts/train_ml.py`.

Table 2: Out-of-sample classification metrics.

Metric	Value
OOS rows	[[RESULT: OOS row count]]
OOS trading days	[[RESULT: OOS trading days]]
ROC-AUC	[[RESULT: OOS ROC-AUC]]
Brier score	[[RESULT: OOS Brier score]]
Base win rate, all rows	[[RESULT: OOS base win rate]]

5.3 Calibration Reliability

Table 3 compares predicted probability buckets with realized win rates. A useful probability model should be monotonic and close to the diagonal, although financial data may remain noisy even after calibration.

Table 3: Reliability table for calibrated probabilities.

Predicted probability bucket	Count	Mean predicted	Actual win rate
[[RESULT: bucket 1 range]]	[[RESULT: bucket 1 n]]	[[RESULT: bucket 1 pred]]	[[RESULT: bucket 1 actual]]
[[RESULT: bucket 2 range]]	[[RESULT: bucket 2 n]]	[[RESULT: bucket 2 pred]]	[[RESULT: bucket 2 actual]]
[[RESULT: bucket 3 range]]	[[RESULT: bucket 3 n]]	[[RESULT: bucket 3 pred]]	[[RESULT: bucket 3 actual]]
[[RESULT: bucket 4 range]]	[[RESULT: bucket 4 n]]	[[RESULT: bucket 4 pred]]	[[RESULT: bucket 4 actual]]
[[RESULT: bucket 5 range]]	[[RESULT: bucket 5 n]]	[[RESULT: bucket 5 pred]]	[[RESULT: bucket 5 actual]]

5.4 Economic Test

The economic evaluation selects top- N setups per day and grades realized R after costs. We report decisive R separately from time exits because unresolved trades marked to the horizon close can otherwise hide weak barrier performance.

Table 4: Baseline comparison for top- N selection after costs.

Selection	Trades	Win/Loss/Timeout	Decisive R	Avg R	PF	Wilson LB
All signals, no selection	[[RESULT: all trades]]	[[RESULT: all W/L/T]]	[[RESULT: all decisive R]]	[[RESULT: all avg R]]	[[RESULT: all PF]]	[[RESULT: all Wilson LB]]
Rule-ranked top- N	[[RESULT: rule trades]]	[[RESULT: rule W/L/T]]	[[RESULT: rule decisive R]]	[[RESULT: rule avg R]]	[[RESULT: rule PF]]	[[RESULT: rule Wilson LB]]
ML-ranked top- N	[[RESULT: ML trades]]	[[RESULT: ML W/L/T]]	[[RESULT: ML decisive R]]	[[RESULT: ML avg R]]	[[RESULT: ML PF]]	[[RESULT: ML Wilson LB]]

The honest interpretation should be filled after running the experiments: `[[RESULT: honest verdict on whether ML beats rule baseline]]`. If the ML-ranked row does not clearly beat the rule-ranked row after costs, the paper should state that result directly.

5.5 Ablations

Table 5 lists required ablations. These are intended to test whether the result depends on calibration, breadth, earnings, and relative or sector strength features.

Table 5: Ablation study.

Variant	ROC-AUC	Brier	Top- N avg R	Top- N decisive R
Full calibrated model	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Uncalibrated GBM	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Drop market/sector breadth	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Drop earnings proximity	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Drop relative/sector strength	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Logistic regression baseline	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]

5.6 Sensitivity Checks

The paper should include sensitivity to top- N , costs, and holding horizon. Table 6 is a placeholder for these results.

Table 6: Sensitivity checks.

Setting	Trades	Avg R	Decisive R	Profit factor
Top-3, cost 0.05R	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Top-5, cost 0.05R	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Top-10, cost 0.05R	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Top-5, cost 0.03R	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Top-5, cost 0.10R	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Hold horizon 3 days	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]
Hold horizon 10 days	[[RESULT]]	[[RESULT]]	[[RESULT]]	[[RESULT]]

5.7 Figures

The final paper should include the following figures. The boxes below are placeholders so the draft compiles before images are generated.

[[FIGURE: calibration / reliability diagram]]
Generate from backend/scripts/train_ml.py OOS predictions.

Figure 1: Predicted probability versus realized win rate.

6 Discussion and Limitations

Several limitations should be kept explicit.

[[FIGURE: distribution of realized R and cumulative R]]
Generate from backend/scripts/ml_oos.pkl.

Figure 2: Realized R distribution and cumulative R for selected trades.

[[FIGURE: sensitivity curve for cost or top-N]]
Generate from top-N and cost sweeps.

Figure 3: Sensitivity of selected-trade R metrics to selection depth or costs.

First, daily bars cannot resolve intraday ordering. If target and stop are both touched within one daily candle, the true first hit is unknown. The offline experiment treats such cases conservatively, but this may still differ from live trade execution.

Second, the universe may contain survivorship and coverage bias. The training set uses symbols available in the cached data. If delisted or formerly liquid stocks are missing, historical performance may be overstated.

Third, the result may be regime dependent. A model that performs only in bullish periods is not necessarily a stock-picking model; it may be a market regime filter. The analysis must therefore report slices by year, direction, and regime.

Fourth, transaction costs are simplified. A flat R-based cost is useful for comparison, but actual execution costs vary by spread, liquidity, slippage, order type, and market conditions.

Fifth, this is not a live trading study. The paper evaluates historical out-of-sample predictions produced by a walk-forward procedure. Live deployment can introduce data delays, corporate-action issues, behavioral selection, and execution errors that are not captured here.

Finally, calibration can drift. The feedback loop records realized outcomes of live signals, but maintaining calibration over time requires regular monitoring. A positive result in one historical sample should be treated as a hypothesis for further validation, not as a permanent property of the market.

7 Conclusion and Future Work

This paper presents a reproducible, leakage-safe framework for evaluating calibrated short-horizon equity setup ranking on NSE stocks. The central claim is methodological: stock-selection systems should be judged with causal features, future-only labels, time-ordered calibration, walk-forward testing, embargoed splits, realistic costs, and transparent baselines. Whether the current model adds a statistically and economically meaningful edge remains to be filled from the experiment outputs. If the result is weak or regime-specific, that is still a useful research outcome because it

prevents overclaiming.

Future work includes adding intraday data to resolve target-stop ordering, testing delisted-symbol coverage, expanding to other markets, comparing more model classes, measuring calibration drift over live paper-trading periods, and formalizing uncertainty intervals for top- N economic metrics.

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Data and Code Availability

The code for the pipeline and experiments is intended to be released at <https://github.com/Kartik962005/ai-trading-mvp>. Daily OHLCV data are obtained through [yfinance](#)/Yahoo Finance subject to their terms of use. Generated caches and experiment outputs should be documented with exact command lines, random seeds, library versions, and train/test split dates before submission.

Educational-Use Disclaimer

This paper is for educational and research purposes only. It is not investment advice, financial advice, or a recommendation to buy, sell, or short any security. Historical backtests and walk-forward evaluations do not guarantee future results.

Numbers to Fill From Experiments

1. Dataset summary: rows, symbols, date range, trading days, exact base win rate. Command: `python scripts/train_ml.py --hold 5 --top-n 5 --cost 0.05`.
2. Main OOS metrics: ROC-AUC, Brier score, OOS row count, OOS trading days. Command: same as above.
3. Reliability table: predicted-probability bucket ranges, counts, mean predicted probability, actual win rate. Command: same as above; copy “Calibration reliability” output.
4. Economic comparison: ML top- N , rule top- N , and all-signals rows: trades, W/L/timeout counts, decisive net R, average net R, profit factor, Wilson lower bound, total R. Command: same as above, with `--top-n 5` and `--top-n 10`.
5. Regime/year/direction slices. Command: run `analyze_ml_oos.py`.
6. Rule-engine baseline report. Command: run `run_backtest.py` with Balanced risk, top-5, cost 0.05R, and hold horizon 5.
7. Sensitivity checks for top- N and cost assumptions. Commands: rerun `train_ml.py` with `--top-n 3`, `--top-n 5`, `--top-n 10`, and costs 0.03, 0.05, and 0.10.

8. Ablation results: calibrated vs uncalibrated, drop breadth, drop earnings, drop relative/sector strength, logistic regression baseline. These may require adding ablation flags/scripts before submission.
9. Figures: calibration diagram, realized R distribution/cumulative R, and top- N or cost sensitivity curve.
10. Final honest verdict: state whether ML-ranked selection beats the rule-ranked baseline out of sample after costs. Do not soften a negative or inconclusive result.

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